Reference Notes

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Contents

Part I

Readings & Literature Review

1. Age-Related Macular Degeneration: A Review

Overview

Age-related macular degeneration (AMD) is a progressive degenerative disease of the macula, the central retinal region responsible for high-acuity vision. AMD is among the leading causes of irreversible vision loss in adults over age 55 and is projected to affect nearly 288 million individuals worldwide by 2040.

The disease primarily affects photoreceptors, retinal pigment epithelium (RPE), bruch membrane, and choriocapillaris. AMD progression involves degeneration of these structures, ultimately leading to central vision loss. Early and intermediate disease may be asymptomatic, while advanced disease can substantially impair reading, driving, facial recognition, and other daily activities.

Pathophysiology

AMD is considered a multifactorial disease driven by interactions among aging, genetic susceptibility, environmental exposures, inflammation, and metabolic dysfunction.

Major biological processes implicated include chronic inflammation, complement pathway dysregulation, oxidative stress, lipid and lipoprotein accumulation, impaired extracellular matrix maintenance and choriocapillaris degeneration. Normal aging causes reduced photoreceptor density lipid deposition within Bruch membrane reduced density of choriocapillaris vessels and increased vascular resistance

These alterations disrupt retinal homeostasis and contribute to extracellular deposit formation. The hallmark lesion of AMD is the druse (plural: drusen), consisting of lipids, proteins, and minerals accumulating between the RPE and Bruch membrane.

Classification of AMD

Stage	Defined By	Clinical Characteristics	OCT Findings	Notes
Early AMD	Medium drusen ($> 63\,\mu m$ and $\leq 125\,\mu m$) No AMD-associated pigmentary abnormalities	Usually asymptomatic Visual acuity often preserved	Small dome-shaped RPE elevations corresponding to drusen Relatively preserved retinal architecture	Progression risk is relatively low at this stage.
Intermediate AMD	Large drusen ($> 125\,\mu m$) And/or pigmentary abnormalities	Difficulty in low-light conditions Reduced contrast sensitivity Metamorphopsia Mild visual blurring	Larger drusen elevations Hyperreflective foci Pigment migration Structural irregularities of the RPE	Most important stage for prediction studies because progression to advanced disease frequently occurs from this state.
Late AMD: Geographic Atrophy	Confluent regions of photoreceptor loss RPE loss Choriocapillaris degeneration	Progressive central vision loss Central scotomas Declining visual acuity	Loss of outer retinal layers RPE attenuation or absence Increased signal transmission into the choroid Hypertransmission defects	Geographic atrophy enlarges over time and is associated with irreversible visual loss.
Late AMD: Neovascular	Growth of abnormal blood vessels beneath or within the retina VEGF-mediated angiogenesis	Often rapid symptom onset Sudden vision decline Metamorphopsia Central visual defects	Intraretinal fluid Subretinal fluid Pigment epithelial detachments Hemorrhage-associated abnormalities	Responsible for most severe acute vision loss associated with AMD.

Table 1: Clinical classification of age-related macular degeneration (AMD).

Major Risk Factors

Age

Age is the strongest risk factor.

The incidence of late AMD increases dramatically with age:

- 0.3 per 1000 individuals aged 55–59
- 5.7 per 1000 individuals aged 75–79
- 36.7 per 1000 individuals aged 90+

Genetics

Estimated heritability is approximately 71%.

Major genetic loci include CFH and ARMS2-HTRA1, which are involved in complement regulation, inflammation, lipid metabolism, and extracellular matrix maintenance.

Smoking

Smoking is the strongest established environmental risk factor. Studies consistently demonstrate substantially increased AMD incidence among heavy smokers.

Diet and Lifestyle

Protective factors include fish consumption, fruits and veggies, and physical activity.

Features Relevant to Progression Prediction

For a progression-prediction project, the review repeatedly highlights several structural features associated with worsening disease.

Drusen Burden

Important characteristics include the number of drusen, total area, volume, and presence of large drusen. Increasing drusen burden characterizes progression from early to intermediate AMD.

Pigmentary Abnormalities

Pigment changes arise from migration of RPE cells into inner retinal layers.

These appear as:

- Hyperpigmentation on fundus imaging
- Hyperreflective foci on OCT

The review identifies pigmentary abnormalities as a major marker of progression risk.

Hyperreflective Foci

Migration of RPE cells creates hyperreflective structures visible on OCT.

These likely reflect active retinal remodeling and degeneration.

Outer Retinal Degeneration

Progressive loss of photoreceptors, ellipsoid zone integrity, and RPE structure may preced advanced disease development.

Hypertransmission Defects

Loss or attenuation of the RPE permits increased OCT signal penetration into deeper tissue.

These appear as vertical transmission columns extending into the choroid.

Such findings are particularly relevant to the current project because the proposed ”barcoding” phenomenon appears visually related to localized transmission abnormalities beneath regions of RPE degeneration.

Geographic Atrophy Precursors

Potential indicators of future atrophy include RPE disruption, hypertransmission, outer retinal thinning, or progressive drusen remodeling.

High-Risk Clinical Profile

The AREDS study reported that:

- 25.9% of individuals with large drusen and pigment abnormalities progressed to late AMD within 5 years
- If one eye already had advanced AMD, progression in the fellow eye reached 53.1% within 5 years when large drusen and pigment abnormalities were present

These findings make large drusen and pigmentary abnormalities particularly important baseline predictors.

2. The Barcode Sign in Optical Coherence Tomography

Overview

El Zawahry et al. described a characteristic OCT imaging pattern termed the *barcode sign*, observed in eyes with extensive corneal neovascularization. The barcode sign appears as multiple vertically oriented dark stripes extending through the OCT image. These stripes arise from attenuation of the OCT signal caused by structures that strongly block light transmission. Although the study focuses on corneal vascularization rather than age-related macular degeneration (AMD), it provides an important example of how biological structures can generate stripe-like OCT patterns through optical shadowing mechanisms.

Study Design

The authors examined five patients with extensive corneal neovascularization arising from:

- Infectious keratitis
- Limbal stem cell deficiency
- Interstitial keratitis

Anterior-segment OCT scans were obtained using a Heidelberg Spectralis OCT system.

For each patient:

- Twenty-one scans were acquired across the superior cornea
- Twenty-one scans were acquired across the inferior cornea
- Forty-two scans were analyzed per eye

OCT findings were compared against slit-lamp examination and clinical imaging.

Description of the Barcode Sign

Major vascular trunks within the cornea completely obstructed OCT light transmission.

This produced:

- A vertical hypo-reflective stripe
- Extension of the stripe posterior to the vessel
- Stripe width proportional to vessel width
- Multiple parallel stripes when several vessels were present

When many vessels were visualized simultaneously, the OCT image resembled a retail barcode pattern, leading to the term "Barcode Sign" where the stripes appeared as dark linear regions embedded within the brighter OCT background.

Proposed Mechanism

The authors hypothesized that the shadowing effect arises primarily from the blood column within actively perfused vessels rather than the vessel wall itself.

Evidence supporting this interpretation includes:

- Both afferent and efferent vessels generated equivalent shadows
- Shadow intensity did not appear dependent on vessel-wall structure
- Ghost vessels lacking active blood flow did not generate shadows

Therefore, the attenuation pattern appears linked to the presence of circulating blood. The barcode sign may thus represent an OCT manifestation of active vascular perfusion.

Relationship to OCT Physics

The barcode sign illustrates a general OCT principle: Structures that strongly attenuate or absorb OCT light can generate vertical shadow artifacts extending into deeper tissue layers.

The resulting pattern consists of:

- Localized signal loss
- Vertical transmission defects
- Stripe-like image structures
- Reduced visibility of deeper anatomy

Such shadowing artifacts are well known in OCT imaging and can reveal information about the underlying structure causing attenuation.

Relevance to AMD Progression Research

Although the barcode sign was identified in corneal vascularization, the study is conceptually relevant to AMD imaging. Several advanced AMD features can alter OCT signal transmission, including:

- Retinal pigment epithelium (RPE) degeneration
- Geographic atrophy
- Hypertransmission defects
- Choroidal structural abnormalities
- Neovascular complexes

These abnormalities can create vertically organized intensity patterns within OCT volumes. Consequently, the barcode sign provides evidence that biologically meaningful stripe-like structures may emerge naturally from tissue-specific optical interactions rather than image-processing artifacts.

Implications for the Current Project

The AMD progression project investigates whether stripe-like "barcoding" patterns visible within OCT scans are associated with future disease progression.

The barcode sign literature suggests several hypotheses:

- Stripe patterns may arise from localized attenuation or transmission phenomena.
- Stripe width may reflect properties of the underlying anatomical structure.
- Stripe density may provide information about disease burden.
- Quantitative characterization of stripe morphology may produce useful imaging biomarkers.

Potential quantitative descriptors include stripe count, spacing, width, contrast, orientation, or spatial frequency content.

While the biological mechanism in AMD may differ from corneal vascular shadowing, this study provides an important precedent demonstrating that barcode-like OCT patterns can arise from meaningful underlying tissue structure and physiology.

3. Self-Attention CNN for Retinal Layer Segmentation

Overview

Cao et al. proposed a hybrid convolutional neural network (CNN) and Transformer architecture for retinal layer segmentation in optical coherence tomography (OCT) images. The primary objective was to improve segmentation accuracy by combining the local feature extraction capabilities of CNNs with the long-range dependency modeling provided by self-attention mechanisms.

The proposed architecture extends the U-Net framework by incorporating:

- A Transformer-based self-attention module,
- Attention Gates (AGs),
- Channel attention mechanisms,
- A one-dimensional attention strategy tailored to retinal OCT anatomy.

The model was evaluated on public retinal OCT datasets and demonstrated improved segmentation performance compared to several existing CNN-based and Transformer-based approaches.

Motivation

Retinal OCT images possess highly structured anatomical organization. Traditional CNNs perform well at extracting local image features but have limited receptive fields, making it difficult to capture long-range spatial relationships.

Challenges in OCT segmentation include layer boundaris spanning large portions of the image, anatomical variability, pathology-induced distortion, and loss of contextual information during downsampling.

The authors argue that successful segmentation requires both local texture information and global structural context. Self-attention mechanisms are introduced to address this limitation.

Self-Attention Mechanism

The Transformer component allows image regions to interact with one another regardless of spatial distance. Self-attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,$$

where Q = query matrix, K = key matrix, V = value matrix, and d = feature dimension.

Unlike convolution, which uses local neighborhoods, self-attention provides a global receptive field. This allows the model to learn relationships between distant retinal structures and improve segmentation consistency across the image.

One-Dimensional Transformer Design

A key innovation of the paper is the use of a one-dimensional Transformer. The authors observed that retinal anatomy is organized primarily along the vertical direction. Instead of computing attention across the entire two-dimensional image, feature maps are rearranged so that attention operates only along the retinal depth dimension.

Feature maps are reshaped from

$$(B, C, H, W) \rightarrow (BW, H, C)$$

where B = batch size, C = number of channels, H = image height, and W = image width.

This design preserves anatomical structure, reduces computational cost, increases effective receptive field, and improves efficiency relative to full 2D attention.

Encoder–Decoder Architecture

The overall architecture follows the encoder–decoder paradigm of U-Net.

Encoder

The encoder consists of repeated blocks of:

- 3×3 convolutions,
- Batch normalization,
- ReLU activation,
- Max pooling.

As depth increases, spatial resolution decreases and feature complexity increases. The encoder extracts increasingly abstract representations of retinal structures.

Decoder

The decoder performs:

- Upsampling,
- Feature fusion,
- Boundary reconstruction,
- Pixel-wise segmentation.

Skip connections transfer high-resolution information from encoder layers to corresponding decoder layers. These connections help preserve fine retinal boundary details that may otherwise be lost during downsampling.

Attention Gates

The model incorporates Attention Gates (AGs) within skip connections. Rather than passing all encoder features directly to the decoder, attention mechanisms determine which features are most relevant.

Conceptually:

$$\tilde{x} = \alpha x, \quad \forall \quad 0 \leq \alpha \leq 1$$

where α is a learned attention coefficient.

Benefits include regularization (suppression) of irrelevant features, improved focus on retinal boundaries, and enhanced segmentation precision.

Channel Attention

In addition to spatial attention, channel attention is used to weight feature channels according to importance. Different channels often capture different types of information:

- Edges,
- Layer boundaries,
- Texture patterns,
- Anatomical structures.

Channel attention allows the network to emphasize informative channels and reduce the influence of less useful features.

Loss Function

Training uses a weighted combination of Dice loss and multiclass cross-entropy loss.

The Dice coefficient is

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}.$$

Corresponding Dice loss L_{Dice} and cross-entropy loss L_{CE} are:

$$L_{\text{Dice}} = 1 - \text{Dice}. \quad L_{\text{CE}} = -\sum_i y_i \log(p_i).$$

The overall objective is

$$L = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{CE}}.$$

Combining the two losses balances pixel level calssification accuracy and global segmentation overlap.

Datasets

The model was evaluated on two retinal OCT datasets:

- DUKE Diabetic Macular Edema (DME) Dataset,
- Optic Disc Retinal OCT Dataset.

These datasets contain manually annotated retinal layer segmentations used as ground truth labels.

Results

Performance was evaluated using the Dice coefficient. For the DUKE DME dataset, an average score of 0.871 was reported, and for the optic-disc dataset, 0.820 was reported. This outperforms UNet, ATtention UNet, TransUNet, RelayNet, and DRUNey. The improvements suggest that combining CNN-based local feature extraction with Transformer-based global context modeling can improve retinal layer segmentation performance.

Key Takeaways

The primary contribution of this work is the integration of self-attention into a U-Net style segmentation framework for retinal OCT images.

Important observations include:

- CNNs effectively learn local retinal texture and boundary information.
- Self-attention provides global contextual information and long-range dependency modeling.
- Restricting attention to the vertical dimension leverages the layered anatomical structure of retinal OCT images.
- Attention Gates improve feature selection during decoding.
- Combining local and global representations improves segmentation accuracy relative to standard CNN architectures.

The paper demonstrates the potential value of hybrid CNN–Transformer architectures for retinal OCT analysis and segmentation tasks.

4. The Barcode Sign in Optical Coherence Tomography

5. Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning

6. Grad-CAM: Visual Explanations from Deep Networks

Part II

Project Directions

7. Classical Statistical Approach for Quantifying Barcoding

Motivation

The observed barcoding phenomenon appears as a vertically oriented transmission artifact within OCT scans. Since the pattern is directional, repetitive, and localized beneath regions of RPE degeneration, it can be treated as a texture-analysis problem rather than an object-detection problem. The primary objective is to convert a visual pattern into a reproducible quantitative biomarker that can later be incorporated into statistical models of disease progression.

Overall Pipeline

The proposed workflow is

$$\text{OCT Scan} \rightarrow \text{Region of Interest} \rightarrow \text{Feature Extraction} \rightarrow \text{Barcode Score}$$

where the region of interest consists of the sub-RPE/choroidal region in which the transmission artifact is observed.

Step 1: Region Selection

The first step is to isolate the anatomical region where the barcode pattern occurs.

Possible approaches include:

- Manual annotation
- Semi-automatic segmentation
- Automated retinal layer segmentation

Restricting analysis to the relevant region reduces noise from unrelated retinal structures.

Step 2: Fourier-Based Barcode Index

For each OCT scan, construct a horizontal transmission profile

$$T(x) = \int_{\text{ROI}} I(x, z)\, dz.$$

The Fourier transform

$$F(f) = \mathcal{F}\{T(x)\}$$

measures spatial frequency content. Periodic stripe patterns should produce elevated spectral energy at characteristic frequencies. A Fourier barcode index can be defined as

$$B_F = \sum_{f \in \mathcal{F}} |F(f)|^2,$$

where $\mathcal{F}$ corresponds to frequencies associated with stripe spacing. Small values indicate weak periodic structure and large values indicate strong stripe organization.

Step 3: Gabor Filter Response

Apply a bank of vertically oriented Gabor filters. A Gabor filter is

$$g(x, y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos(2\pi f x').$$

The filter response is

$$R(x, y) = (I * g)(x, y).$$

Compute

$$B_G = \frac{1}{N} \sum_{x,y} |R(x, y)|.$$

This quantity measures how strongly the image resembles an oriented stripe pattern.

Step 4: Directionality via Structure Tensor

The local structure tensor is

$$S = G_\sigma * \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}.$$

Let $\lambda_1 \geq \lambda_2$ be the eigenvalues. Define anisotropy as

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}.$$

Interpretation: $A \approx 0$: isotropic texture, $A \approx 1$: highly directional pattern. Strong vertical transmission artifacts should exhibit elevated anisotropy.

Composite Barcode Score

The final score can combine multiple measurements:

$$S_i = w_1 B_{\text{Fourier}} + w_2 B_{\text{Gabor}} + w_3 A + w_4 C,$$

where A = anisotropy score and C = local contrast measure. The weights can be estimated statistically or chosen based on clinical interpretation. This is easy to interpret and works well with rsmaller datasets. It is reliant on imaging physics and produces a continuous biomarker for downstream analysis.

8. Deep Learning Approach for Quantifying Barcoding

Motivation

Classical image-processing methods require manual feature engineering. Deep learning instead learns image representations directly from OCT scans. Rather than specifying stripe characteristics explicitly, the network discovers image features most useful for quantifying barcoding.

Overall Pipeline

The proposed workflow is

$$\text{OCT Scan} \rightarrow \text{CNN Encoder} \rightarrow \text{Deep Features} \rightarrow \text{Barcode Severity Score.}$$

The output may be a continuous barcode severity score, a segmentation mask, or a heatmap identifying barcode regions.

Encoder Network

A ResNet-style encoder may be used. Convolutional layers compute

$$Y(i,j) = \sum_m \sum_n K(m,n) X(i+m, j+n).$$

Where early layers learn edges, gradients, and local texture, and deeper layers learn retinal structures, transmission adnormalities, and disease related patterns.

Regression-Based Severity Scoring

The final network output is $\hat{S}_i = f_\theta(X_i)$, where X_i is the OCT image, $\hat{S}_i$ is predicted barcode severity. Training minimizes

$$L = \frac{1}{n} \sum_i (S_i - \hat{S}_i)^2,$$

where S_i represents clinician-assigned severity labels.

Segmentation-Based Approach

An alternative approach is to train a U-Net with an output $M(x,y) \in [0,1]$ representing the probability that pixel (x,y) belongs to a barcode-like region.

We'll use Dice loss:

$$L_{\text{Dice}} = 1 - \frac{2|M \cap G|}{|M| + |G|},$$

where G denotes the expert annotation. The overall barcode score may then be

$$S_i = \int M(x, y)\, dx\, dy.$$

Self-Attention Extensions

Because OCT anatomy is highly structured, attention mechanisms may improve performance. Self-attention computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

This gives us larger receptive fields, better use of anatomical concepts, and improved identification of subtle artifacts.

Interpretability

Grad-CAM can identify image regions driving predictions. Importance weights are

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k}.$$

Heatmaps are generated as

$$L = \text{ReLU}\left(\sum_k \alpha_k A^k\right).$$

This provides visual confirmation that the model is using barcode-like structures rather than unrelated image regions. This learns features automatically and has no information loss, leading to higher accuracy than crafted features.

9. Binary Progression Prediction Model

Objective

The goal is to determine whether barcode measurements obtained from a baseline OCT scan can predict future AMD progression. The outcome is binary:

$$Y_i = \begin{cases} 1, & \text{rapid progression,} \\ 0, & \text{slow or no progression.} \end{cases}$$

The exact definition of progression depends on available data. Possible endpoints include conversion to geographic atrophy or rapid geographic enlargement.

Predictor Variables

Predictors may include our previously defined barcoding score, along with demographic covariates like age, and other known biomarkers.

Logistic Regression Model

A natural starting point is logistic regression:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 S_i + \beta_2 X_{i1} + \cdots + \beta_p X_{ip},$$

where$p_i = P(Y_i = 1)$ and e^{β_1} represents the odds ratio associated with a one-unit increase in barcode score.

Model Evaluation

Performance can be assessed using AUC and confusion matrices, along with calibration curves.

Clinical Question

The key comparison is whether or not adding the barcode score improves prediction significantly.

10. Bayesian Progression Prediction Model

Motivation

Barcode measurements are inherently uncertain. Different image-processing methods, scanners, or graders may produce slightly different values. A Bayesian model explicitly accounts for this uncertainty.

Latent Barcode Severity

Instead of treating the barcode score as fixed, define Z_i = latent barcode severity.

Observed image measurements are noisy observations:

$$X_i \mid Z_i \sim N(Z_i, \sigma_X^2).$$

The latent severity becomes the true biological quantity of interest.

Progression Model

Let

$$Y_i = \begin{cases} 1, & \text{progression,} \\ 0, & \text{no progression.} \end{cases}$$

Then $Y_i \mid Z_i \sim \text{Bernoulli}(p_i)$, with $\text{logit}(p_i) = \alpha + \beta Z_i + \gamma^\top C_i$, where C_i contains clinical covariates.

Prior Distributions

Example priors:

$$\alpha \sim N(0, 10^2) \qquad \beta \sim N(0, 10^2) \qquad Z_i \sim N(\mu_Z, \sigma_Z^2)$$

The posterior distribution becomes $p(\alpha, \beta, Z \mid X, Y)$. We can then sample with MCMC methods or variational inference.

Outputs

For each patient, the model produces:

$$P(\text{progression} \mid \text{OCT}),$$

along with a full posterior distribution, which yields mean risk estimates, credible intervals, and better uncertainty quantification. This ensures image uncertainty gets incorporated, and also seperates measurement noise from biological signal.

Part III

Notes

11. Statistical Methods for Barcode Quantification

Fourier Analysis

The Fourier transform decomposes an image into spatial frequency components and is one of the most natural tools for analyzing repetitive stripe-like patterns.

Let $I(x, y)$ denote an OCT image. Its two-dimensional Fourier transform is

$$F(u, v) = \sum_x \sum_y I(x, y) e^{-2\pi i(ux+vy)}.$$

The corresponding power spectrum is

$$P(u, v) = |F(u, v)|^2.$$

Interpretation:

- Low frequencies correspond to broad intensity variations.
- High frequencies correspond to rapidly changing structures.
- Periodic stripe patterns generate concentrated spectral energy at characteristic frequencies.

For barcode-like transmission artifacts, a common approach is to collapse the choroidal region into a one-dimensional signal

$$I(x) = \int_{\text{choroid}} I(x, z)\, dz,$$

and then analyze its frequency content.

If stripe spacing is approximately d then its dominant frequency is $f = \frac{1}{d}$

A possible Barcoding Index is

$$B = \sum_{f \in \mathcal{F}} |F(f)|^2,$$

where $\mathcal{F}$ is a frequency band corresponding to stripe spacing.

This measure is highly interpretable, computationally efficient, and quantifies periodicity well. The trade-off being it assumes approximate uniform spacing and therefore loses its spatial localization.

Gabor Filters

Gabor filters are localized frequency detectors that simultaneously measure orientation, frequency, and spatial location. A two-dimensional Gabor filter is

$$g(x, y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right)\cos(2\pi f x' + \phi),$$

where

$$x' = x\cos\theta + y\sin\theta, \qquad y' = -y\sin\theta + x\cos\theta.$$

and θ = orientation, f = frequency, σ = scale, γ = aspect ratio, ϕ = phase shift. The response of the filter is

$$R(x, y) = (I * g)(x, y).$$

Strong responses occur when image structures match the filter orientation and frequency.

For OCT barcoding:

- Vertical filters isolate transmission columns.
- Multiple frequencies allow detection of varying stripe widths.
- Average filter response can serve as a barcode score.

Unlike Fourier analysis, Gabor filters preserve spatial localization.

Structure Tensor

The structure tensor is a classical method for quantifying local orientation.

First compute image gradients

$$I_x = \frac{\partial I}{\partial x}, \qquad I_y = \frac{\partial I}{\partial y}.$$

The local structure tensor is

$$J = \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}.$$

In practice, local averaging is performed:

$$S = G_\sigma * J,$$

where G_σ is a Gaussian smoothing kernel.

Let λ_i be the eigenvalues of S.

Interpretation:

- $\lambda_1 \approx \lambda_2$: isotropic texture.
- $\lambda_1 \gg \lambda_2$: strong directional structure.

The dominant eigenvector indicates the primary orientation of local image patterns. For OCT images, structure tensors quantify the strength and orientation of vertical transmission artifacts.

Anisotropy Metrics

Anisotropy measures how strongly image structure is aligned along a preferred direction.

A common metric is

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}, \forall \quad A \in [0, 1]$$

Interpretation:

- $A = 0$: completely isotropic texture. (weak directional organization)
- $A \approx 1$: highly directional pattern. (strong vertical stripe structure)

An anisotropy score is attractive because it is simple and scale independent, and may correlate with disease severity.

Latent Variable Bayesian Models

One challenge is that "barcoding" is not a directly observed biological structure. Instead, it may represent a latent imaging phenomenon associated with underlying degeneration. Let Z_i be latent barcode severity for eye i.

Observed image features$X_i = (B_i, G_i, A_i, \ldots)$ and may include:

- Fourier-based barcode scores
- Gabor filter responses
- Anisotropy measurements
- Texture features

A hierarchical model can be written as

$$X_i \mid Z_i \sim p(X_i \mid Z_i), \qquad Y_i \mid Z_i \sim p(Y_i \mid Z_i),$$

where Y_i is the progression outcome. Posterior inference is based on

$$p(Z_i \mid X_i, Y_i) = \frac{p(Y_i \mid Z_i) p(X_i \mid Z_i) p(Z_i)}{p(X_i, Y_i)}.$$

Advantages:

- Separates measurement noise from biological signal.
- Produces uncertainty estimates.
- Allows probabilistic risk prediction.

Rather than classifying scans as barcoding present or absent, the model estimates a posterior distribution over severity.

Using Barcode Scores as Predictors

The ultimate objective is progression prediction. Suppose S_i is a barcode score computed from image features. A logistic regression model can be written as

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 S_i + \beta_2 X_{i1} + \cdots + \beta_p X_{ip},$$

where p_i is the probability of preogression.

For time-to-event outcomes, a Cox model may be used:

$$h(t \mid X) = h_0(t)\exp(\beta^T X).$$

Important questions include:

- Does the barcode score predict progression?
- Does it improve prediction beyond existing AMD biomarkers?
- Is the score reproducible across scanners and readers?

Ultimately, barcode quantification is useful only if it provides clinically meaningful prognostic information.

12. CNNs for OCT Analysis

Convolution Layers

The core operation of a CNN is convolution.

Given an image $X \in \mathbb{R}^{H\times W}$ and a filter $K \in \mathbb{R}^{k\times k}$, the convolution operation is

$$Y(i,j) = \sum_m \sum_n K(m,n) X(i+m, j+n).$$

Each filter learns a feature detector. Early CNN layers typically learn edges, gradients, and simple textures, whereas deeper layers learn anatomical structures, retinal layers, and disease-related patterns. Important hyperparameters are kernel size, stride, zero-padding, and number of filters, and output dimensions satisfy:

$$W_{\text{out}} = \frac{W - F + 2P}{S} + 1.$$

Pooling Layers

Pooling reduces spatial dimensionality.

Max pooling computes

$$Y(i,j) = \max_{(m,n)\in\mathcal{N}} X(i+m, j+n).$$

Average pooling computes

$$Y(i,j) = \frac{1}{|\mathcal{N}|} \sum_{(m,n)\in\mathcal{N}} X(i+m, j+n).$$

Benefits:

- Reduced computation and overfitting
- Increased translation invariance

Typical pooling windows are either 2×2 or 3×3.

Activation Functions

Activation functions introduce nonlinearity.

ReLU: $f(x) = \max(0, x)$.

Leaky ReLU: $f(x) = \max(\alpha x, x)$.

Sigmoid: $f(x) = \frac{1}{1+e^{-x}}$.

Tanh: $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

ReLU is dominant in modern CNNs because it mitigates vanishing gradients and is computationally efficient.

Fully Connected Layers

After convolutional feature extraction, feature maps are flattened into a vector $x \in \mathbb{R}^d$. A fully connected layer computes

$$y = Wx + b.$$

Classification probabilities are obtained using softmax:

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}.$$

For binary classification:

$$p = \frac{1}{1+e^{-z}}.$$

Fully connected layers combine learned image features into disease predictions.

CNN Feature Learning

A key advantage of CNNs is automatic feature learning. CNNs learn feature representations directly from data.

The representation learned by layer ℓ can be written as $h_\ell = f_\ell(h_{\ell-1})$. Successive layers learn increasingly abstract image representations:

$$\text{Edges} \to \text{Textures} \to \text{Shapes} \to \text{Pathology}.$$

ResNet

Deep networks suffer from vanishing gradients. ResNet introduces residual connections:

$$y = F(x) + x.$$

The identity shortcut allows gradients to flow through many layers. This is also easier to optimize and has better convergence, and most modern OCT classifiers use the ResNet backbone (EX: ResNet-18/34/50/101).

U-Net

U-Net is the dominant architecture for biomedical image segmentation.

Architecture:

- Encoder (downsampling)
- Bottleneck
- Decoder (upsampling)
- Skip connections

The encoder extracts context while the decoder restores spatial resolution. Skip connections preserve fine anatomical detail.

Transfer Learning

Medical imaging datasets are often small. Transfer learning uses a model pretrained on a large dataset (e.g., ImageNet) and adapts it to a new task.

Let f_θ denote a pretrained CNN. Training proceeds by:

1. Initializing weights from the pretrained model.
2. Replacing the final classification layer.
3. Fine-tuning on OCT images.

This has lower training time and generalizes better, and performs better on small datasets.

Interpretable CNNs (Grad-CAM)

A common criticism of CNNs is their lack of interpretability. Gradient-weighted Class Activation Mapping (Grad-CAM) identifies image regions responsible for predictions. Let A^k denote feature map k. Importance weights are computed as

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k},$$

where y is the model output. The Grad-CAM heatmap is

$$L = \mathrm{ReLU}\left(\sum_k \alpha_k A^k \right).$$

The resulting map highlights image regions most influential for the prediction. In OCT analysis, Grad-CAM can help determine whether the network is utilizing:

- RPE abnormalities
- Hypertransmission regions
- Barcode-like artifacts
- Other retinal structures

thereby improving interpretability and clinical trust.

13. Hyperparameter Tuning and Extension Toward Localization

13.1. Tuning ResNet50 for Binary Barcoding Classification

The OCT-C8 model provides a pretrained OCT feature extractor that can be adapted to binary barcoding detection. The new classification task is

$$y \in \{0, 1\},$$

where 0 denotes barcode-negative and 1 denotes barcode-positive.

The first architectural change is replacing the eight-class output layer with a two-class output layer. The convolutional backbone can initially be retained from the OCT-C8 fine-tuned model.

Key hyperparameters to tune include:

- **Learning rate:** controls the size of parameter updates during training. Candidate values can be tested on a log scale, such as $\eta \in \{10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$.
- **Fine-tuning depth:** determines which ResNet layers are unfrozen. Candidate settings include $d \in \{0, 1, 2, 3\}$, where $d = 0$ trains only the classifier, $d = 1$ unfreezes Layer 4, $d = 2$

unfreezes Layers 3–4, and $d = 3$ unfreezes Layers 2–4.

- **Batch size:** controls how many images are processed before each optimization step. Candidate values include $B \in \{8, 16, 32, 64\}$, depending on GPU memory.
- **Weight decay:** regularizes model weights to reduce overfitting. Candidate values can be tested on a log scale, such as $\lambda \in \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$.
- **Dropout:** may be added before the final classifier to reduce overfitting. Candidate probabilities include $p_{\text{drop}} \in \{0, 0.1, 0.2, 0.3, 0.5\}$.
- **Class weighting:** addresses imbalance between barcode-positive and barcode-negative scans. Candidate positive-class weights can be tested as $w_+ \in \{1, 2, 3, 5, 10\}$, or defined from class counts using $w_+ = N_-/N_+$.
- **Data augmentation:** improves robustness to scan variability. Candidate augmentation strengths can be indexed as $a \in \{0, 1, 2, 3\}$, where $a = 0$ uses no augmentation, $a = 1$ uses light flips/crops, $a = 2$ adds mild brightness and contrast jitter, and $a = 3$ adds stronger geometric and intensity perturbations.
- **Decision threshold:** converts predicted probability into a positive or negative classification. Candidate thresholds can be tested over $\tau \in [0.10, 0.90]$ with step size $\Delta\tau = 0.05$.

A reasonable tuning sequence is:

1. Train only the final classifier with the ResNet backbone frozen.
2. Unfreeze Layer 4 and fine-tune with a small learning rate.
3. If the dataset is sufficiently large, unfreeze Layers 3 and 4.
4. Compare performance across learning rates such as 10^{-3}, 10^{-4}, and 10^{-5}.
5. Tune the probability threshold for barcode-positive classification based on sensitivity, specificity, and clinical priorities.

Because missing true barcoding cases may be clinically undesirable, threshold selection should not rely only on accuracy. Instead, candidate thresholds should be evaluated using:

- **Sensitivity (Recall):** measures the proportion of barcode-positive scans correctly identified by the model.

  Sensitivity $= \frac{TP}{TP+FN}$
- **Positive Predictive Value (PPV, Precision):** measures the probability that a scan predicted as barcode-positive is truly positive.

  PPV $= \frac{TP}{TP+FP}$
- **Precision-Recall AUC (PR-AUC):** summarizes the tradeoff between precision and recall across all decision thresholds.

Let

Precision$(\tau) = \frac{TP(\tau)}{TP(\tau)+FP(\tau)}$

and

Recall$(\tau) = \frac{TP(\tau)}{TP(\tau)+FN(\tau)}$.

Then

PR-AUC $= \int_0^1 \text{Precision}(\text{Recall})\, d(\text{Recall})$.

- **Expected Calibration Error (ECE):** measures agreement between predicted probabilities and observed frequencies.

  ECE $= \sum_{m=1}^{M} \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$

  where B_m denotes the mth probability bin, acc(B_m) is the empirical accuracy within the bin, and conf(B_m) is the average predicted probability within the bin.

For barcoding detection, sensitivity is prioritized to minimize missed barcode-positive scans, PPV quantifies confidence in positive detections, PR-AUC is preferred over ROC-AUC under class imbalance, and ECE evaluates whether predicted probabilities can be interpreted as reliable confidence estimates.

If barcode-positive cases are rare, precision-recall AUC may be more informative than overall accuracy.

13.2. Tuning Grad-CAM for Barcoding Localization

Grad-CAM does not have trainable parameters in the same way as ResNet50, but it does involve design choices that affect interpretability.

Important Grad-CAM choices include:

- **Target layer:** the convolutional layer used to generate the heatmap.
- **Target class:** barcode-positive probability should be used for localization.
- **Heatmap threshold:** determines which activated pixels are treated as candidate barcode regions.
- **Smoothing:** reduces noisy activation patterns.
- **Minimum region size:** removes small isolated heatmap artifacts.

The default target layer is the final ResNet block, usually Layer 4. This gives semantically meaningful localization but spatially coarse maps. If localization is too diffuse, earlier layers such as Layer 3 can be tested because they retain higher spatial resolution.

A practical approach is to evaluate multiple Grad-CAM variants:

- Layer 4 Grad-CAM,

- Layer 3 Grad-CAM,
- Smoothed Grad-CAM,
- Thresholded Grad-CAM,
- Grad-CAM combined with classical texture measures.

The key validation question is not only whether the classifier is correct, but whether the heatmap overlaps with the clinically suspected barcode or hypertransmission region.

13.3. From Classification to Bounding Boxes

Once binary classification is reliable, the next step is localization. The simplest approach is to use Grad-CAM heatmaps as weak localization maps.

A candidate bounding box can be generated by:

1. Compute the Grad-CAM heatmap for the barcode-positive class.
2. Threshold the heatmap, for example retaining the top 10–20% of activation values.
3. Remove small isolated regions.
4. Identify the largest connected activated region.
5. Draw the smallest rectangle enclosing that region.

This produces an approximate barcode ROI without requiring pixel-level segmentation labels.

The bounding box can be written as

$$(x_{\min}, y_{\min}, x_{\max}, y_{\max}),$$

where $x_{\min}$ and $x_{\max}$ define the horizontal extent and $y_{\min}$ and $y_{\max}$ define the vertical extent.

If clinician-provided bounding boxes become available, the model can shift from weak localization to supervised object detection using architectures such as Faster R-CNN, RetinaNet, or YOLO.

At that stage, hyperparameters shift from classification tuning to detection tuning, including:

- Anchor box sizes,
- Anchor aspect ratios,
- Intersection-over-union threshold,
- Non-maximum suppression threshold,
- Detection confidence threshold,
- Bounding-box regression loss weight.

The primary evaluation metric also changes. Instead of accuracy or AUC, localization should be evaluated using intersection-over-union (IoU), defined as

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}.$$

A predicted box is considered accurate if it sufficiently overlaps the clinician-defined barcode region.

13.4. From Bounding Boxes to Pixel-Level Measurements

After the barcode region is localized, the bounding box becomes the analysis ROI. Within this ROI, a second-stage measurement pipeline can estimate barcode morphology.

Candidate measurements include:

- Total barcode width,
- Total barcode height,
- Bounding-box area,
- Number of individual bars,
- Mean bar width,
- Median bar width,
- Minimum and maximum bar width,
- Bar density,
- Fraction of ROI occupied by barcode-positive pixels.

This shifts the model output from a probability,

$$P(\text{barcoding present}),$$

to quantitative measurements of barcode extent and morphology.

13.5. Pixel-to-Micron Conversion

Pixel-based measurements are useful for algorithm development, but clinical interpretation requires physical units.

If OCT metadata provide scale factors, let s_x denote microns per pixel horizontally and s_y denote microns per pixel vertically.

A pixel width w_{px} can be converted into microns by

$$w_{\mu m} = w_{\text{px}} s_x.$$

Similarly, a pixel height h_{px} becomes

$$h_{\mu m} = h_{\mathrm{px}} s_y.$$

Area conversion is given by

$$A_{\mu m^2} = A_{\mathrm{px}} s_x s_y.$$

Therefore, once barcode-positive regions are localized, the same measurements can be reported in clinically interpretable units such as microns or square microns.

This requires preserving scan metadata from the original OCT volume files, especially pixel spacing and acquisition geometry.

13.6. Overall Development Path

The intended development path is:

C8 OCT Classification $\rightarrow$ Binary Barcoding Classification $\rightarrow$ Grad-CAM Localization $\rightarrow$ Bounding Box Detection

At each stage, the tuning target changes:

- Classification stage: optimize discrimination and calibration.
- Grad-CAM stage: optimize clinical plausibility of localization.
- Bounding-box stage: optimize overlap with clinician-defined ROIs.
- Measurement stage: optimize repeatability and agreement with expert assessment.
- Physical-scale stage: ensure measurements are comparable across scans and devices.

This staged approach preserves the strengths of the current CNN framework while gradually moving toward clinically interpretable barcoding measurements.

Part IV

Trials

14. 6/7: Initial Testing

This is the first attempt at quantifying the barcoding artifact that we observed in the images. The longer goal is to be able to quantify the severity of this artifact and then assess the robustness across scans and patients, and use a comparison study to see if it can qualify as a good biomarker or feature for downstream analysis.

14.1. Feature Exploration

We want to see whether the barcoding phenomenon could be represented quantitatively using simple image-derived features. Three representative scans exhibiting best-case barcoding were selected for exploratory analysis. All images were converted to grayscale and intensity-normalized to place measurements on a common scale.

The barcoding artifact appears as vertically oriented regions of increased transmission extending into the choroid, so analysis was restricted to a manually selected region of interest (ROI) containing the RPE/choroid interface and the upper choroid. Several candidate crops were visually evaluated. A crop spanning pixel rows 180–350 was selected because it retained the visible transmission columns while excluding most of the vitreous, neurosensory retina, and low-information background signal.

As a first proof-of-concept approach, the 2D ROI was reduced to a 1D transmission profile. For each horizontal image position, the average pixel intensity within the ROI was computed. Conceptually, this profile measures the amount of light transmitted through the retina into the underlying choroid. Areas of RPE disruption are expected to exhibit increased transmission and therefore appear as brighter regions within the profile.

Initial inspection revealed a substantial large-scale brightness gradient across the scan, reflecting global imaging and anatomical effects rather than the barcoding artifact itself. To isolate local transmission variations, a heavily smoothed trend estimate was subtracted from the transmission profile, producing a detrended signal centered approximately around zero. This detrending step allowed local fluctuations associated with transmission columns to be examined independently of overall image brightness.

Several simple summary statistics were then extracted from the detrended profile, including the SD, variance, and number of prominent local peaks. For the representative examples, SD ranged from 0.0200 to 0.0325, variance ranged from 0.000398 to 0.001057, and peak counts ranged from 24 to 27. Although these measurements demonstrated that the barcoding pattern could be converted into quantitative features, differences between scans were relatively modest.

Importantly, the resulting numerical rankings did not fully align with visual assessments of barcoding severity. Images that appeared to exhibit prominent barcoding did not consistently produce the largest transmission-profile statistics. This suggests that maybe reducing the OCT image to a 1D signal discards information that may be important for characterizing the artifact.

Clinically, barcoding is not defined solely by intensity variation. The phenomenon is also characterized by its vertical orientation, localized appearance, and irregular spacing within the choroid. These properties are not explicitly represented by transmission-profile statistics. While the transmission-profile approach provided a useful proof of concept, the results suggested that methods preserving 2D information would likely be required for more effective quantification of the artifact.

This finding motivated subsequent investigation of texture-analysis approaches, including Gabor filtering and directional image features designed specifically to quantify vertically oriented structures.

14.2. Directional Texture Quantification Using Gabor Filters

The next objective was to develop a quantitative feature that explicitly captures the vertical texture pattern observed in OCT barcoding.

We employ a Gabor filter, a commonly used image-analysis tool designed to detect structures with a specific orientation and spacing. Conceptually, a Gabor filter functions as a directional texture detector. Rather than measuring overall image brightness, it measures how strongly an image contains repetitive structures aligned in a particular direction. Because the barcoding artifact appears clinically as a series of vertically oriented transmission columns extending into the choroid, Gabor filtering provides a natural framework for quantification.

Two key parameters determine the behavior of a Gabor filter. Orientation, which specifies the direction of the texture being detected, and spatial frequency, which determines the characteristic spacing between repeating structures. To identify the most informative parameter settings, a systematic grid search was performed across multiple frequencies (0.02–0.20) and orientations (0 to $\pi/2$ radians). The analysis was restricted to the same ROI selected in the previous notebook.

Frequency analysis demonstrated that the strongest responses generally occurred at low frequencies. For the representative example used during initial testing, the peak response occurred near a frequency of 0.04, corresponding to an estimated spacing of approximately 25 pixels between adjacent transmission columns. This finding suggests that the barcoding artifact is composed primarily of relatively broad structures rather than fine-scale image texture.

Orientation analysis revealed a clear directional preference. The strongest response was observed at an orientation of $\pi/2$, corresponding to vertical image structures. This result is consistent with clinical observations.

To assess reproducibility, the frequency-orientation search was repeated for all three representative scans. The highest observed responses were:

- amd_sample1: 0.0141 (frequency = 0.02, orientation = $\pi/2$)
- amd_sample3: 0.0048 (frequency = 0.02, orientation = $\pi/2$)
- amd_sample2: 0.0028 (frequency = 0.06, orientation = 0)

Interestingly, Samples 1 and 3 demonstrated similar response patterns, with maximal responses occurring at low frequencies and vertical orientations. Sample 2 exhibited a different response structure, suggesting either weaker barcoding, greater structural heterogeneity, or the presence of competing image features influencing the measurement.

Because ROI selection may influence texture measurements, a sensitivity analysis was performed using progressively deeper choroidal crops. Although response magnitude decreased as additional low-information pixels were included, the overall ranking of scans remained stable. For example, the Gabor score for Sample 1 decreased from 0.0141 to 0.0045 as the ROI was expanded, while maintaining the highest response among the representative examples. This behavior suggests that the measurement is influenced by ROI definition but remains directionally consistent across reasonable crop choices.

Overall, these findings indicate that Gabor-based texture analysis captures important structural information that is lost when OCT images are reduced to 1D transmission profiles. The method successfully identifies the directional nature of the barcoding artifact and provides an interpretable quantitative measure of transmission-column prominence. These results motivated further investigation of complementary directional texture measures, including structural tensor analysis and anisotropy-based metrics.

14.3. Directional Organization and Anisotropy Analysis

A second image-analysis approach was investigated to determine whether the barcoding phenomenon exhibits measurable directional organization. Structure tensor analysis evaluates the local orientation of image features directly. This provides a complementary method for quantifying the degree of organization present within the transmission-column pattern.

The structure tensor is a commonly used image-processing technique that identifies whether local image regions contain a preferred direction. Regions containing aligned structures, such as vessels, fibers, or transmission columns, produce stronger directional signals than regions with random texture. To summarize this directional organization, an anisotropy score was computed. Anisotropy measures the extent to which image features are oriented in a common direction. Higher values indicate greater directional organization, whereas lower values indicate more isotropic or randomly oriented texture.

Initial analysis of the full ROI demonstrated that anisotropy measurements were influenced by several anatomical structures, including retinal layer boundaries, the foveal depression, the RPE interface, and the transmission columns themselves. To better isolate the barcoding pattern, the analysis was repeated using a barcode-focused ROI restricted to the upper choroid, where the transmission columns were most visually apparent. This reduced the influence of unrelated retinal anatomy while preserving the structures of interest. The mean anisotropy score decreased only slightly from 0.451 in the full ROI to 0.430 in the barcode-focused ROI, suggesting that directional organization remained prominent after restricting attention to the choroidal region.

The barcode-focused analysis was then applied to all three representative scans. The resulting anisotropy scores were:

- amd_sample1: 0.430
- amd_sample3: 0.382
- amd_sample2: 0.341

The ranking is consistent with Gabor filter analysis, which is important because the two approaches measure different image characteristics. Gabor filters quantify oriented texture patterns at specific spatial scales, whereas anisotropy measures the overall directional organization of image features without assuming a particular spacing between structures. The consistency between these methods suggests that both are detecting a genuine property of the barcoding pattern rather than an artifact of a particular algorithm.

From a clinical perspective, the anisotropy results support the hypothesis that barcoding is not simply a collection of isolated transmission defects. Instead, the phenomenon appears to possess coherent directional organization, with stronger examples exhibiting more pronounced alignment of transmission columns. The ability of anisotropy measurements to distinguish among the representative scans provides further evidence that directional image features may serve as useful quantitative biomarkers.

14.4. Development of a Composite Barcoding Index

The previous analyses demonstrated that multiple image-derived features capture different aspects of the barcoding phenomenon. Transmission-profile statistics quantify overall signal variability, Gabor filtering measures directional texture strength, and anisotropy analysis quantifies directional organization. Because no single feature fully characterizes the artifact, the next objective was to investigate whether these complementary measurements could be combined into a unified quantitative score.

To construct a prototype barcoding index, four features were incorporated:

- Transmission-profile standard deviation,
- Peak count from the detrended transmission profile,
- Gabor texture score,
- Anisotropy score.

Because these measurements exist on different numerical scales, each feature was normalized relative to its maximum observed value before aggregation. An initial composite index was then constructed using equal weighting of all four features.

The resulting composite scores were:

- amd_sample1: 0.945
- amd_sample3: 0.808
- amd_sample2: 0.624

This produces the same ranking, so thd ordering was consistent with both visual inspection and the rankings obtained from the Gabor and anisotropy analyses. The agreement suggests that multiple independent image characteristics contribute to the appearance of barcoding and can be integrated into a single quantitative measure.

Because the exploratory analyses indicated that texture and directional organization appeared more informative than simple transmission-profile statistics, a second texture-oriented index was evaluated. In this formulation, greater weight was assigned to the Gabor and anisotropy features while standard deviation and peak count received smaller contributions. Specifically,

$$BI_{\text{texture}} = 0.4G + 0.4A + 0.1S + 0.1P,$$

where G denotes the normalized Gabor score, A the normalized anisotropy score, S the normalized standard deviation, and P the normalized peak count.

The resulting weighted scores were:

- amd_sample1: 0.978
- amd_sample3: 0.692
- amd_sample2: 0.547

Importantly, the ranking remained unchanged:

$$\text{Image 1} > \text{Image 3} > \text{Image 2}.$$

The stability of the ranking under different weighting schemes suggests that the observed ordering is not driven by a single feature or modeling choice. Rather, multiple independent measurements identify the same image as exhibiting the strongest apparent barcoding and the same image as exhibiting the weakest.

The present index should be viewed as a proof-of-concept biomarker rather than a finalized clinical severity scale. The analysis is limited to three representative OCT scans, ROI selection remains manually defined, and the feature weights were selected heuristically rather than learned from outcome data.

14.5. Robustness Check

Before such measurements can be considered useful biomarkers, it is important to evaluate their stability under reasonable variations in image processing choices.

The most important user-defined parameter in the current workflow is ROI selection. Because the transmission columns are not sharply bounded anatomical structures, small differences in crop placement may occur during manual image review. To assess the sensitivity of the proposed measurements to ROI definition, the vertical crop boundaries were systematically shifted while recomputing both the Gabor texture score and the anisotropy score.

Five ROI configurations were evaluated, spanning pixel rows 170–340 through 210–380. These perturbations were selected to represent realistic variations in manual ROI placement while maintaining focus on the RPE/choroid complex and the upper choroid.

Across all three representative scans, both the Gabor and anisotropy measurements decreased gradually as the ROI was shifted deeper into the choroid. This behavior is expected because the strongest transmission-column signal is concentrated near the RPE/choroid interface. As progressively deeper regions were included, the proportion of low-information tissue increased, resulting in lower overall feature values.

Despite these changes in magnitude, the relative ordering of the representative scans remained unchanged throughout the analysis. This finding is particularly important because the primary objective of the proposed barcoding index is not necessarily to assign an absolute severity value, but rather to distinguish among scans exhibiting different degrees of the phenomenon. The magnitude of variation was modest. For example, the mean anisotropy score ranged from 0.420 for Sample 1 to 0.340 for Sample 2, with standard deviations of 0.017 and 0.006, respectively.

Formal bootstrap-based uncertainty estimation was initially considered. However, repeated computation of Gabor and structure tensor features proved computationally expensive during this exploratory phase of development. Consequently, ROI perturbation analysis was adopted as a practical first assessment of robustness.

15. 6/13: Barcoding CNN, a Three-Stage Sketch

The initial barcoding index and image-processing analyses demonstrated that the barcoding phenomenon appears to possess measurable image characteristics. However, these approaches rely on manually defined ROIs and handcrafted features. An alternative direction is to use a convolutional neural network (CNN) to identify and localize barcoding directly from OCT images.

15.1. CNN for Barcoding Detection

The first stage would be to identify an existing CNN-based object detection or segmentation framework from the OCT literature and adapt it to the barcoding problem. Candidate architectures include Faster R-CNN, RetinaNet, YOLO, or OCT-specific segmentation networks. The model would be trained using hyptertransmission AMD OCT scans labeled as barcoding-positive or barcoding-negative. The primary output would be:

- Presence or absence of barcoding,
- Confidence score for barcoding detection.

This stage addresses the basic question: *Can a CNN reliably distinguish scans exhibiting barcoding from scans without barcoding?*

15.2. CNN for Bounding Barcoding

Once detection performance is satisfactory, the second stage would focus on localization. Rather than simply classifying a scan, the model would be trained to identify a bounding box containing the barcode-positive region. This would require clinician-provided annotations indicating the approximate location of hypertransmission or barcoding within each scan.

15.3. CNN for Pixel-to-Meter Conversion

A third stage would focus on extracting quantitative measurements from the detected ROI. Using the image scale information available within the OCT scan, pixel measurements could be converted into physical units. Candidate measurements include:

- Overall barcode width, length, and area
- Sufficient statistics: Number of bars, Mean bar width, Median bar width, Minimum and maximum bar width,
- Bar density and fraction of the ROI occupied by barcode-positive tissue.

These measurements could then be incorporated into future progression-risk models and evaluated alongside conventional AMD biomarkers.

15.4. CNN for OCT Classification Results

Before developing a CNN specifically for barcoding detection, a proof-of-concept study was conducted to determine whether a modern convolutional neural network could learn clinically meaningful OCT image features and provide interpretable localization of retinal pathology. A publicly available OCT dataset containing 24,000 images across eight retinal conditions (AMD, CNV, CSR, DME, DR, DRUSEN, MH, and NORMAL) was used for this purpose.

Initial exploratory analysis confirmed that the dataset was balanced across disease classes, with 3,000 images per condition. Images exhibited varying spatial resolutions, with widths ranging from 512 to 1536 pixels and heights ranging from 496 to 586 pixels. Both grayscale and RGB images were present, so we converted everything to RGB and scaled everything to 224x224.

A ResNet50 architecture pretrained on ImageNet was selected as the baseline model. Transfer learning was used because retinal OCT datasets are substantially smaller than the datasets typically required to train deep convolutional networks from scratch. Two training strategies were evaluated.

The first strategy froze the pretrained convolutional backbone and trained only the final classification layer. After three epochs, the model achieved approximately 85% validation accuracy and 86% test accuracy. Performance varied across disease categories, with AMD achieving near-perfect classification while DRUSEN and NORMAL exhibited greater confusion. Although classification performance was encouraging, the resulting Grad-CAM visualizations occasionally highlighted broad regions of the image, suggesting that the model's learned representations were still relatively coarse.

To improve OCT-specific feature learning, a second model was trained by fine-tuning the final residual block (Layer 4) together with the classification head while keeping earlier layers frozen. This approach substantially improved performance. The fine-tuned model achieved a test accuracy of 93.2%, with corresponding class-specific performance summarized below:

Table 2: Classification performance of the fine-tuned ResNet50 model on the OCT test set.

Class	Precision	Recall	F1 Score
AMD	1.00	1.00	1.00
CNV	0.88	0.85	0.86
CSR	1.00	1.00	1.00
DME	0.94	0.85	0.89
DR	1.00	0.99	0.99
DRUSEN	0.82	0.82	0.82
MH	0.99	1.00	0.99
NORMAL	0.84	0.95	0.89

The improvement is below:

Table 3: Overall classification performance of the ResNet50 models.

Model	Test Accuracy
Frozen Backbone	0.86
Layer 4 Fine-Tuned	0.932

The strongest performance was observed for AMD, CSR, DR, and MH, all of which achieved near-perfect classification. DRUSEN remained the most challenging category, which is clinically reasonable given the overlap between drusen morphology, early AMD features, and normal retinal appearance.

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to both the frozen and fine-tuned models. Grad-CAM produces heatmaps that identify image regions contributing most strongly to a particular prediction. The frozen model produced relatively diffuse activation maps, whereas the fine-tuned model consistently localized attention to retinal structures associated with disease pathology.

For AMD and DRUSEN images, activation regions frequently concentrated near the outer retina, retinal pigment epithelium (RPE), and choroidal interface, which are the anatomical locations where disease-related abnormalities are expected. For NORMAL scans, activations were concentrated around the foveal contour and preserved retinal layer architecture. Importantly, the model did not appear to rely primarily on image borders, scanner artifacts, or irrelevant background structures.

Although Grad-CAM does not provide pixel-level segmentation, these results suggest that the CNN learned biologically plausible retinal representations and was capable of localizing clinically meaningful image features. This proof-of-concept provides evidence that CNN-based approaches may be suitable for future barcoding detection and localization tasks. In particular, the ability of the model to identify disease-associated retinal regions supports the feasibility of adapting similar architectures for barcode-positive versus barcode-negative classification, bounding-box localization of hypertransmission regions, and subsequent extraction of quantitative barcode measurements.